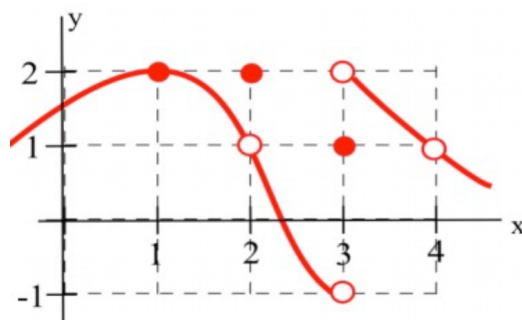


HW 5.2.1: Limits

Graphical and Algebraic Techniques

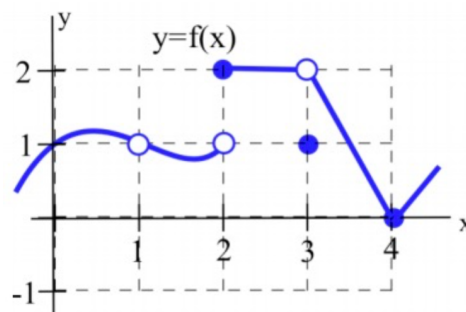
1. Use the figure to the right to answer the following

- a. $\lim_{x \rightarrow 2} f(x) =$
- b. $f(2) =$
- c. $\lim_{x \rightarrow 3^+} f(x) =$
- d. $\lim_{x \rightarrow 3^-} f(x) =$
- e. $\lim_{x \rightarrow 3} f(x) =$
- f. $f(3) =$
- g. $\lim_{x \rightarrow 4} f(x) =$



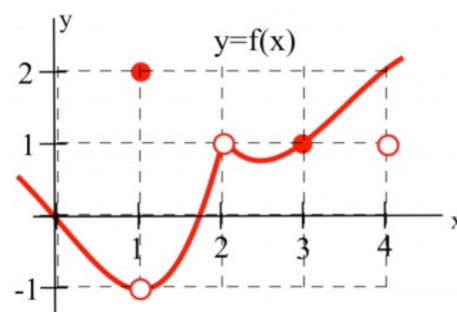
2. Use the figure to the right to answer the following

- a. $\lim_{x \rightarrow 1} f(x) =$
- b. $\lim_{x \rightarrow 2^-} f(x) =$
- c. $\lim_{x \rightarrow 2^+} f(x) =$
- d. $\lim_{x \rightarrow 2} f(x) =$
- e. $f(2) =$
- f. $\lim_{x \rightarrow 3} f(x) =$
- g. $f(3) =$



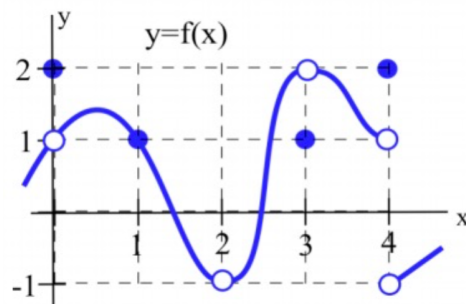
3. Use the figure to the right to answer the following

- a. $\lim_{x \rightarrow 1} f(x) =$
- b. $f(1) =$
- c. $\lim_{x \rightarrow 2} f(x) =$
- d. $f(2) =$
- e. $\lim_{x \rightarrow 3} f(x) =$
- f. $f(3) =$



4. Use the figure to the right to answer the following

- a. $\lim_{x \rightarrow 0} f(x) =$
- b. $f(0) =$
- c. $\lim_{x \rightarrow 2} f(x) =$
- d. $f(2) =$
- e. $\lim_{x \rightarrow 4} f(x) =$
- f. $f(4) =$



Find the limit if it exists

5. $\lim_{x \rightarrow 1} \frac{x^2 + 3x + 2}{x - 5}$

6. $\lim_{x \rightarrow 5} \frac{x^2 + 3x + 2}{x - 5}$

7. $\lim_{x \rightarrow 1} \frac{x^2 - x - 2}{x - 1}$

8. $\lim_{x \rightarrow 0} \frac{x + 9}{x^2 + 11x + 18}$

9. $\lim_{x \rightarrow -9} \frac{x + 9}{x^2 + 11x + 18}$

10. $\lim_{x \rightarrow -2} \frac{x + 9}{x^2 + 11x + 18}$

11. $\lim_{x \rightarrow 0} \frac{\frac{1}{x+1} - 1}{x}$

12. $\lim_{x \rightarrow 0} \frac{\frac{1}{2-x} - \frac{1}{2}}{x}$

13. $\lim_{x \rightarrow -4} \frac{x}{\frac{1}{4+x} - \frac{1}{4}}$

14. $\lim_{x \rightarrow -2} \frac{\frac{1}{2} + \frac{1}{x}}{2+x}$

15. $\lim_{x \rightarrow \infty} \frac{2x^2 - 5x - 3}{x^2 + 5x + 6}$

16. $\lim_{x \rightarrow \infty} \frac{5x^2 - 4x - 1}{x^3 + 2}$

Selected Answers:

1. a. 1 b. 2 c. 2 d. -1 e. DNE f. 1 g. 1
3. a. -1 b. 2 c. 1 d. undefined e. 1 f. 1
5. $-\frac{3}{2}$
7. DNE
9. $-\frac{1}{7}$
11. -1
13. 0
15. 2